Constant-acceleration problem-solving

LEARNING TARGET · R5 I can choose and use an equation for motion with steady acceleration.

A · KEY VOCABULARY

Steady velocity	Equal gaps; velocity does not change. Use v × t.
Constant acceleration	Velocity changes by the same amount each second.
Average velocity	(start + final) ÷ 2 — only when acceleration is constant.
Method	A choice with a reason: "I chose ___ because the velocity ___."

B · KEY EQUATIONS + UNITS

$$v = v_0 + a\,t$$

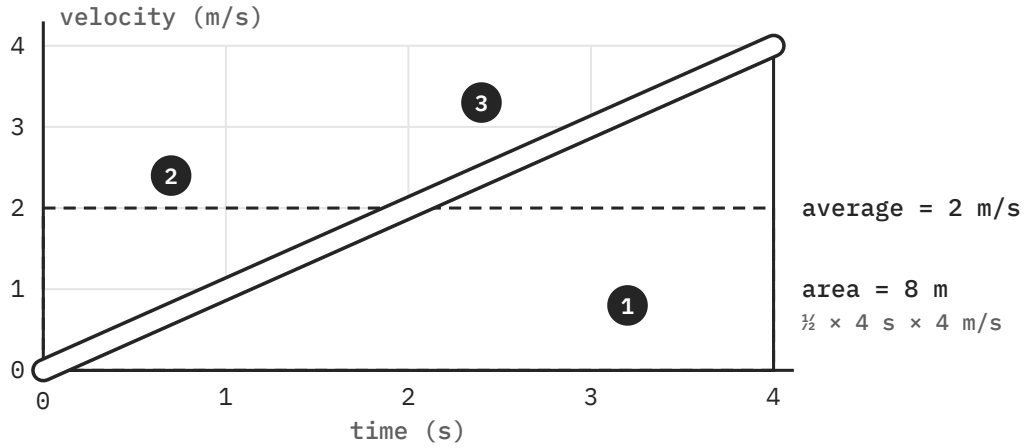
$$\Delta x = v_0 t + \tfrac{1}{2} a t^2$$

$$v^2 = v_0^2 + 2a\Delta x \quad \text{(no t)}$$

1 · 2 list given · list asked
3 · 4 choose · check units + sign

C · COLOR IT · TRIANGLE OR AVERAGE: SAME AREA

0 → 4 m/s in 4 s, constant a.



☐ **1 · BLUE** Triangle area = displacement

☐ **2 · PURPLE** Average-velocity box (dashed)

☐ **3 · PURPLE** Velocity line

D · WORKED EXAMPLE

A cart speeds up steadily from 0 to 8 m/s in 4 s. Find its displacement.

GIVEN	EQUATION	WORK	ANSWER
$v_i = 0$, $v_f = 8$ m/s $t = 4$ s constant a	$\Delta x = \frac{1}{2}(v_i + v_f) \times t$	average = (0 + 8) ÷ 2 = 4 m/s $\Delta x = 4 \times 4$	**+16 m** average works: a is constant

E · THE 2026-XJ THREAD

DAY 8 OF 22 · NEW TOOL: CHOOSING A MODEL

Steady velocity or steady acceleration? Choosing the right model for XJ is the difference between a forecast that holds and one that drifts.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else